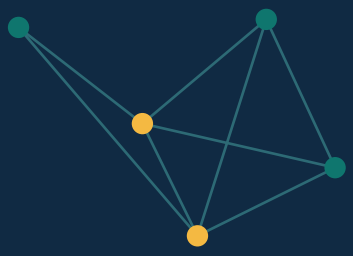
FINAL INNOVATION CONCEPT

SukariSol

One connected digital intelligence platform for Kenya's sugar value chain



MAP • COORDINATE • RECONCILE

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Themes: Climate-smart agronomy; regional competitiveness; financing and investment; out-grower institutions; consumer protection and market diversification; AI and industrial automation.

Abstract

Kenya's sugar value chain links farmers, out-grower institutions, transporters, mills, distributors and consumers, yet decisions can be slowed by fragmented records and limited shared visibility. SukariSol proposes an integrated digital platform connecting geospatial farm intelligence, agronomy support, harvest logistics, delivery and payment reconciliation, factory reliability, traceability, financing-readiness insights, by-product opportunities and market planning. Its micro-apps share governed records, role-based access and audit trails while allowing phased adoption. Satellite-derived vegetation indicators will support monitoring, not be treated as direct yield measurements. The project will begin with stakeholder research and data validation, then demonstrate a farm-to-delivery-to-reconciliation workflow with a willing industry partner. Benefits—including reduced waiting, clearer records and better planning—will be treated as hypotheses and assessed against agreed baselines during a pilot.

Keywords: sugarcane; digital agriculture; geospatial intelligence; logistics; traceability; industrial reliability; circular economy

1. The opportunity

Kenya's sugar industry faces connected operational challenges including delayed harvesting and cane lifting, delayed payments in some mills, maintenance delays and poor feeder-road conditions, as documented by the Kenya National Bureau of Statistics [2]. These are linked problems: crop readiness affects harvest plans; harvest plans affect truck allocation and mill intake; delivery records inform farmer accounts; factory output and stock influence distribution. SukariSol proposes a shared digital operating layer that connects authorized information to practical decisions rather than replacing every existing system.

The 2026 Kenya Sugar Board innovation call welcomes practical concepts with a credible pathway to application [1]. Existing services, including Miwa Bora, will be reviewed for complementarity and possible integration rather than assumed absent or inferior [3].

CONNECTED WORKFLOW

FARM Map & monitor	→	HARVEST Plan & dispatch	→	MILL Receive	→	PAYMENT Reconcile	→	MARKET Plan & sell

2. One platform, a family of micro-apps

Six core operational modules anchor the first release. Supporting micro-apps broaden the long-term vision; their implementation will be prioritized after user research, feasibility checks and partner validation.

CaneYield / CaneVision

Register farm boundaries and acreage; map locations and road distances to mills/markets; display satellite observations and vegetation indicators; combine with crop stage and harvest history. Satellite proxies are not direct yield measurements.

CaneDoctor

Crop-stage reminders, agronomy guidance, pest/disease reporting, weather-aware advice and links to extension services and input providers, with local validation.

CaneFlow

Coordinate harvest requests, readiness, truck capacity, route options, dispatch, mill receiving slots and delivery events. Start with transparent rules and baseline comparisons.

FairCane / CaneTrust

Connect authorized weighbridge and delivery records to price rules, itemized deductions, farmer statements, payment status and dispute workflows. Flag issues for review, not automatic blame.

MillMind

Equipment registers, inspections, faults, repair histories, downtime and spare parts. Begin with rules and alerts; predictive maintenance only if reliable data exists.

CaneCycle / ByProduct Hub

List eligible bagasse, molasses, filter cake and other materials; match verified listings to buyers after accounting for existing uses, contracts, quality, handling, storage, transport and rules.

SugarMarket Intel / SugarMarket

Use authorized sales, orders, inventory and transport records for demand scenarios, stock visibility, distribution planning and market intelligence. Validate forecasts before use.

SugarShield

Traceability across approved farm, delivery, processing and distribution events, including quality/lot events where data is available.

ShambaScore / SukariSol Capital

Explore consent-based, explainable farm and transaction profiles that may inform financing readiness. Not an automatic credit decision; financial use requires partner, fairness, privacy and regulatory review.

OutGrower OS

Grower registration, contracts/consents, field visits, input distribution, extension records, harvest planning, communications and organization reporting.

Miwa Mkononi

Farmer-facing mobile experience for alerts, farm information, delivery/payment status, agronomy resources and support, with low-bandwidth and offline-friendly capture where feasible.

SukariSol Command Center

Role-specific dashboards for mills, grower organizations, logistics teams and other authorized users, with cross-module indicators, exceptions and action queues.

Shared foundation: identity and organization records; role-based permissions; consent; audit trails; notifications; import/export; documented APIs; reporting; and responsive web/mobile interfaces. Start with a modular monolith and geospatial-capable relational database; separate services only when justified.

3. Implementation and first demonstrator

MONTH 1	**Discover**	Interview farmers, out-growers, transporters, agronomists, mill operations/maintenance, finance teams and distributors. Map workflows, pain points, decision owners and pilot interest.
MONTH 2	**Test feasibility**	Inventory farm, harvest, weighbridge, payment, maintenance, sales and stock data. Test mapping/satellite observations; assess data rights, cloud cover, road-data quality and integration constraints. Review Miwa Bora and other existing services.
MONTH 3	**Validate scope**	Test prototypes with intended users. Prioritize by problem severity, value, data readiness, feasibility and differentiation. Agree pilot scope, permissions, baseline metrics and evaluation.
MONTH 4	**Build connected core**	Implement shared identity, farm records, geospatial interface, basic satellite observations, harvest scheduling, delivery records and payment reconciliation. Add other dashboards where data permits.
MONTH 5	**Prepare pilot**	Test access controls, calculations, audit trails, mobile usability, data quality and integrations. Prepare documentation, training and a pilot report template.

FIRST DEMONSTRATOR

Register and map a farm → record crop stage and estimated harvest window → create collection order → assign truck → record mill delivery → reconcile delivery against authorized payment records → show status to the right users. Other capabilities may begin as clearly labelled mock-ups; the project will distinguish functional features, simulations and planned work.

Evaluation indicators

Module	Measures to test
CaneYield / CaneVision	Validated farm boundaries; estimation error where reliable ground data exists.
CaneFlow	Truck waiting time, empty travel, punctuality, request-to-collection time.
FairCane / CaneTrust	Delivery reconciliation rate; time to resolve discrepancies.
MillMind	Recorded downtime; maintenance response; history completeness.
CaneCycle / ByProduct Hub	Qualified enquiries/transactions; indicative net value after costs.
SugarMarket / SugarShield	Forecast error; stock imbalance; traceability completeness.
Farmer & grower tools	Adoption, task completion, user clarity and record completeness.

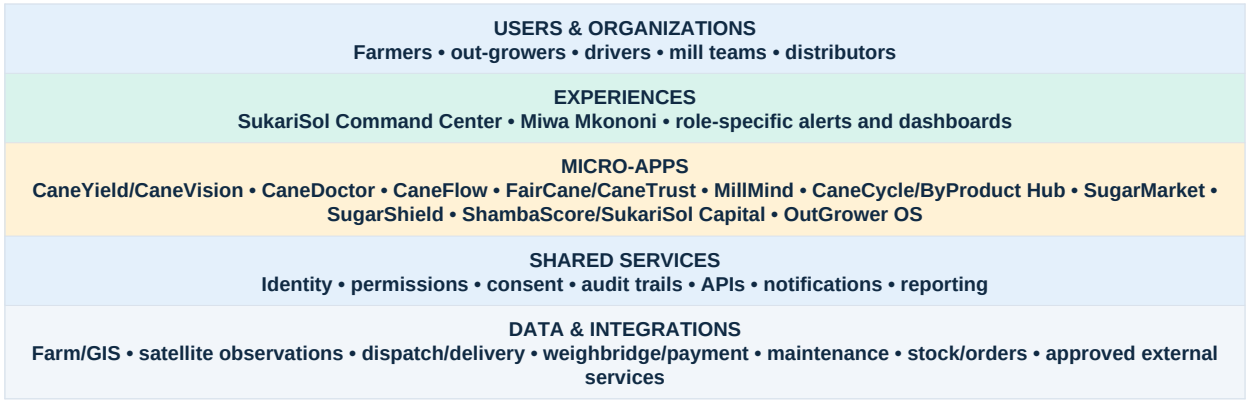
Risks and safeguards: incomplete data, inaccurate estimates, integration limits, connectivity and adoption. Use field validation, uncertainty labels, training, offline-friendly workflows where feasible, least-privilege access, consent and audit logs. Farmer and factory records will not be universally visible.

4. Adoption, safeguards and scale

Initial customers are likely to be sugar mills and out-grower institutions, with transport contractors, distributors and selected service providers as additional users. Potential revenue includes a mill-funded subscription or enterprise licence, implementation and data migration, training, integrations and optional advanced analytics once their value is validated. Willingness to pay and the operating model will be tested in stakeholder research.

The first pilot should involve one willing mill or out-grower organization and a defined group of farms and transport workflows. It should establish a baseline, use agreed data-sharing arrangements and test whether the connected workflow improves visibility or measurable outcomes without disrupting operations. Expansion should depend on results, user feedback, data quality and a sustainable support model.

Platform architecture



Data principles: collect only what is needed; clarify ownership and permitted use; protect personal and commercially sensitive records; document calculations; keep audit history; and make uncertainty visible. Financing-readiness tools require consent, explainability and review before lender use. By-product listings must respect existing uses, contracts, quality requirements and applicable rules.

5. Conclusion

SukariSol aims to connect decisions shaping sugar production, movement, processing, farmer transactions and market access through a governed digital platform. It combines six operational pillars with supporting agronomy, grower management, traceability, finance-readiness and farmer-facing tools. Rather than claiming every feature is new or already built, SukariSol will validate existing solutions, secure a pilot partner and demonstrate one measurable end-to-end workflow before scaling. Success will be judged by trusted records, adoption and evidence of operational value.

References

[1] Kenya Sugar Board. “Call for Innovations 2026,” Sugar Industry Innovation Symposium & Expo. https://sugarinnovation.go.ke/call-for-papers/

[2] Kenya National Bureau of Statistics. *National Agriculture Production Report 2024*, section “Challenges facing the Sugar Industry.” https://www.knbs.or.ke/wp-content/uploads/2025/01/National-Agriculture-Production-Report-2024.pdf

[3] Agriculture and Food Authority. “Miwa Bora Mobile App.” https://www.afa.go.ke/miwa-bora-mobile-app/

[4] European Space Agency. “Sentinel-2.” https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-2

[5] Google Earth Engine. “Earth Engine Documentation.” https://developers.google.com/earth-engine

Declaration: SukariSol is a proposed concept and implementation pathway, not a report of a completed system or measured pilot results. Projected benefits remain to be tested. Authors should confirm compliance with the current call’s rules and portal requirements.